BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

PHYSICS (10th) — (Fresh/Re-appear) — (Version-A) — SECTION –A

Time Allowed: 20 Min | **Maximum Marks:** 12 | **Paper Code:** BN-XXXII-I

Note: Note: There are THREE sections in this paper i.e A, B & C. Attempt section A and return it to the superintendent within the given time. Q.No.1: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct fill that circle in front of the question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question.

Q1. The bending of waves around the edges of the obstacle is.

(A) Reflection
(B) Refraction
(C) Diffraction
(D) Damping

Q2. The speed of sound on a warm day when the outdoor temperature is 38°C is

(A) 331 m/s
(B) 354 m/s
(C) 345 m/s
(D) 362 m/s

Q3. Which of the following cannot transmit sound?

(A) Solid
(B) Liquid
(C) Gas
(D) Vaccum

Q4. A convex lens with focal length 8.00 cm has the power of lens

(A) 2.05 D
(B) 4 D
(C) 12.5 D
(D) 16 D

Q5. The unit of Electric Potential is

(A) Farad
(B) Coulomb
(C) Volt
(D) NC^{-1}

Q6. A capacitor C has a charge Q. The actual charges on its plates are.

(A) Q, Q
(B) Q,0
(C) Q/2, Q/2
(D) Q, -Q

Q7. The wire made from which of the following material is a conductor.

(A) Silver
(B) Silk
(C) Glass
(D) Rubber

Q8. Slip rings are part of

(A) DC motor
(B) AC generator
(C) Transformer
(D) Magnet

Q9. A step-up transformer increases

(A) Power
(B) Voltage
(C) Energy
(D) Current

Q10. The Boolean equation for NAND gate is

(A) X = AB
(B) X = A+B
(C) X = AB with overline
(D) X = A with overline + B with overline

Q11. Telephone communication is the example of transmission of signals through

(A) Electric wires
(B) Optical fibers
(C) Electromagnetic waves
(D) All

Q12. Origin of energy from the sun and star is

(A) Fusion
(B) Fission
(C) Carbon dating
(D) Radioactivity

PHYSICS (10th) — (Fresh/Re-appear) — SECTION –B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 32

> Note: Note: Attempt section B & C accordingly.

Q2. (Marks 32) Q.No 2. Attempt any EIGHT parts of the following. All parts carry equal marks.

i. Differentiate between transverse and longitudinal wave.

ii. Define and explain intensity of sound waves.

iii. Why sound cannot be heard on moon?

iv. Define linear magnification. Write its mathematical form.

v. What is capacitor? Define capacitance and its units.

vi. What is electric potential? In what units we measure electric potential?

vii. Calculate the resistance of a 40W automobile headlight designed for 12V.

viii. Defing magnetic field also write its S.I unit.

ix. Write the uses of cathode ray oscilloscope.

x. Can internet be used for shopping? Give examples.

xi. If nuclear radiation is harmful, how it can be used for treatment of diseases?

SECTION –C

Maximum Marks: 21

Note: Note: Attempt any THREE of the following. All questions carry equal marks

Q3.

a) Define simple Harmonic motion. Show that mass spring system executes simple harmonic motion.

b) A sound wave of wavelength 1.7×10^{-2} m. Calculate the frequency of sound if its velocity is 343.4 m/s.

Q4.

a) What are optical fibers? Write its three applications.

b) What is speed of light in water having index of refraction as 1.33?

Q5.

a) State and explain Ohm's law. What are its limitations?

b) A small electric heater has a resistance of 15 ohms, when the current in it is 2 amperes. What voltage is required to produce this current?

Q6.

a) Define logic gates. Describe the operation of AND, OR and NOR gates by drawing their symbol and truth tables.

b) Write three uses of Radioisotopes.